

Planar versus embedded PbS quantum-dot integration in perovskite solar cells

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This work covers two approaches of integrating PbS quantum dots (PbS QDs) into perovskite-based solar cells. We compare a planar PbS quantum-dot film (D1) against 5 nm PbS dots embedded in MAPbI₃ via a Maxwell-Garnett effective medium (D2) in a common FTO/SnO₂/MAPbI₃/PbSQD/ p-PbS-EDT/MoO_x/Au stack. Both devices have been studied using coupled 3D finite-difference time-domain optics and 2D drift-diffusion transport. In an unfiltered limit, in which every absorbed photon is counted as generating a collectable carrier, the optimized D1 and D2 devices reach 24.00% and 24.21% power conversion efficiency (PCE), respectively. Enforcing spectral consistency, so that only above-bandgap absorption generates carriers, the planar cell leads by 4.57 percentage points (18.80% versus 14.23%). This ranking remains unchanged when the quantum-dot volume is matched at equal active thickness. The embedded cell reaches parity only if sub-bandgap collection is fully operative, which a homogeneous single-gap model cannot represent. We conclude that planar integration is the more robust route under conventional single-gap transport, while the embedded architecture depends on a sub-bandgap collection channel that has not been demonstrated at the loading and in the morphology modelled here.

I. INTRODUCTION

Due to rising global energy demand, reliance on fossil fuels has grown; although they remain a key pillar of economic development, their impact on climate change and human health is urging us to find a better, cleaner and more sustainable alternative. At the centre of that transition lies Photovoltaics (PV), which involves the conversion of sunlight into usable electrical energy in devices commonly known as solar cells (SCs). The sun provides more than 120,000 terawatts, more than 10,000 times the current global energy demand, which promotes solar as one of the best renewable energy sources.

Perovskite Solar cells (PSCs) are a third-generation solar-cell technology that has emerged rapidly as a leading technology in the PV industry and research. Certified single-junction perovskite devices now exceed 26% PCE [17], comparable to well established silicon technologies while offering lower manufacturing costs and easier synthesis, making perovskite cells one of the best candidates for renewable energy. Metal-halide perovskite cells are known for their high absorption coefficients and long carrier-transport lengths, with electron-hole diffusion lengths exceeding one micrometre in solution-processed films [6], which enable them to collect large amounts of the solar photons while using thin-film active layers. One drawback is that any photon with energy below the bandgap of the PSC passes through it or is converted into heat in a layer other than the active layer, an energy loss that could instead be used to generate electricity. For a 1.55 eV MAPbI₃ absorber this cutoff falls at 800 nm, so the entire near-infrared portion of the solar spectrum is unavailable to a bare perovskite absorber. To meet the world's increasing demand for energy, further enhancements to perovskite light harvesting and charge collection are a major research area.

Quantum Dots are semiconducting nanoparticles that

can be used as light absorbers, with a band gap tunable through the particle size and surface chemistry by quantum confinement [15]. Lead-sulfide QDs (PbS QDs) are one of the leading materials in quantum dot solar cells (QDSCs); they act as a low-bandgap absorber that helps the PSC by capturing low-energy photons and thereby increasing the cell efficiency, but they also introduce side effects such as trap-assisted recombination and electronic disorder. Standalone PbS QD cells have reached 13.62% certified efficiency [8], with comparable values reported for QD-ink and heterojunction routes [9, 10], while the contribution collected beyond 1100 nm remains on the order of 1% [12]. To maximize the efficiency of QDSCs we must therefore tune the QDs such that the added carrier-collection benefits outweigh the drawbacks they introduce [4, 5].

Two integration strategies of QDSCs are particularly relevant. In the first, a stacked-layer approach, the perovskite layer remains distinct from the QD layer; each layer can then be configured separately, with its own thickness, bandgap and electrical parameters, which simplifies the carrier-pathway description, but it creates a heterointerface between the perovskite and the QD layer that makes the device prone to voltage and fill-factor reductions from interface defects. Stacked quantum-dot architectures of this kind are well established, with bilayer all-perovskite-QD cells reaching 15.6% [11]. In the second, the QDs are embedded inside the perovskite layer itself, which removes the discrete planar heterojunction and increases the optical overlap of the two materials, but at the cost of a more complicated carrier pathway: the QDs are isolated particles rather than a continuous layer, so carrier transfer and confinement can no longer be represented by a simple bulk material with simple parameters. Experimentally, PbS-in-perovskite hybrids have reached 18.6%, with the improvement attributed to enhanced perovskite crystallisation, current

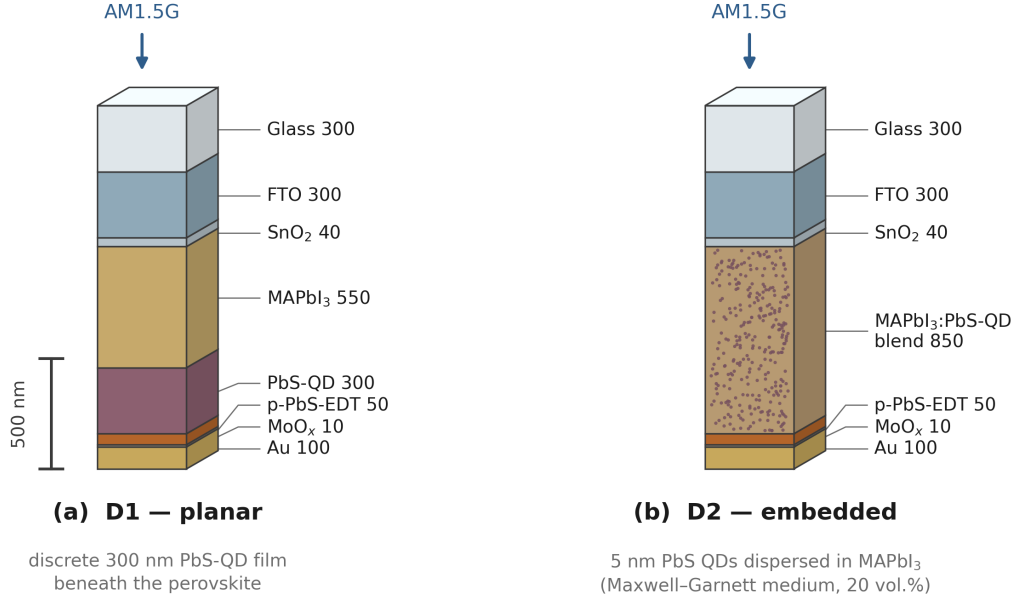


FIG. 1. Device architectures compared in this work. The contacts, transport layers, illumination and simulation workflow are identical, and the active layer is 850 nm thick in both cells; the design question is the placement of the quantum dots. (a) D1 confines the dots to a discrete 300 nm PbS-QD film beneath the perovskite, introducing an internal MAPbI₃/QD heterointerface. (b) D2 disperses 5 nm PbS dots throughout the 850 nm absorber, represented optically by a Maxwell-Garnett effective medium at 20 vol.% and removing that discrete junction. The two active layers also differ in dot volume, dot-phase continuity, assumed collection edge, and electrical representation, as set out in Sec. II A and controlled for in Sec. V A. Layer thicknesses are to scale and labelled in nm; lateral extent and perspective depth are schematic.

TABLE I. Summary of device performance under the two collection limits. Unfiltered values assume every absorbed photon is collected; strict single-gap values are obtained by spectrally filtering the generation profile and re-solving the drift-diffusion problem in full. The high-voltage row is a separate doping-driven operating point rather than a variant of the optimized morphology comparison. All efficiencies are referenced to 100 mW cm⁻².

Collection limit	Device	J_{sc} (mA cm ⁻²)	V_{oc} (V)	FF	PCE (%)
Unfiltered	D1 optimized	42.65	0.741	0.759	24.00
Unfiltered	D2 optimized	50.17	0.719	0.671	24.21
Unfiltered	D2, $p = 10^{16}$ cm ⁻³	36.92	0.915	0.737	24.91
Strict single-gap	D1 optimized	33.86	0.729	0.762	18.80
Strict single-gap	D2 optimized	29.69	0.691	0.694	14.23
Strict single-gap	D2, $p = 10^{16}$ cm ⁻³	21.81	0.909	0.745	14.78

density and fill factor rather than to infrared harvesting [3]. Sub-bandgap photocurrent has nonetheless been demonstrated directly in this material system, both as an operating intermediate-band device [1] and through photoconductivity evidence for a delocalized mini-band [2].

Existing simulation studies of such hybrids are predominantly one-dimensional, using drift-diffusion tools such as SCAPS-1D that do not resolve the full-wave optical response of the stack [13, 14]. Light management by embedded inclusions has been examined by the present group in the complementary resonant regime, where ellipsoidal plasmonic TiN nanoparticles dispersed in MAPbI₃ enhance absorption through a geometry-tunable local-

ized surface plasmon resonance [28]. The present work addresses the opposite limit, in which the inclusions are non-resonant dielectric quantum dots well below the wavelength, and in which the design levers are consequently different. To our knowledge the planar-versus-embedded question has not been addressed with three-dimensional full-wave optics coupled to drift-diffusion transport.

This work compares a planar PbS-QD/MAPbI₃ device with a device with embedded PbS QDs in MAPbI₃, using the same contacts, device-layer structure and illumination spectrum within a coupled optical and electrical framework. The aims are to find out how much additional

current the embedded device generates when we assume perfect carrier collection, to answer whether removing the internal interface makes the embedded device less prone to recombination and processing variability, and to separate the optical enhancements from the assumption that each photon is converted and collected as a carrier.

II. SIMULATION AND METHODOLOGY

A. Device Architectures and Material Stack

The two devices share the same overall structure (Fig. 1), such that the front to back layering consists of: Glass (300 nm)/FTO (300 nm)/SnO₂ (40 nm)/composite active layer/PbS-EDT (50 nm)/MoO_x (10 nm)/Au (100 nm), for a total stack thickness of 1650 nm. The composite active layer is the only part of the stack that differs between the two devices, where in D1 it consists of a 550 nm layer of MAPbI₃ and a separate 300 nm PbS QD layer, while for D2 it consists of one 850 nm active blend layer of MAPbI₃ with 5 nm PbS QD dispersed in the perovskite layer. Within that layer the two devices differ in more than the placement of the dots: they also differ in dot volume, in the continuity of the dot phase, in the collection edge of the absorber (1.03 eV for the discrete film against 1.44 eV for the blend surrogate), and in their electrical representation. These differences are quantified where they affect the comparison, in Sec. V A for loading and in Sec. V for the collection edge. The composite layer thickness is therefore unified across the two devices at 850 nm to allow a fair comparison between the devices' performance. The baseline D2 device uses a 20% dot fraction while a 25% fraction is used in the grating-optimized case after the parameter sweep. D1 treats the QD layer as a low bandgap continuous semiconductor with specific optical and charge transport parameters; this approximation represents a ligand-exchanged PbS material with a 1.03 eV bandgap. D2 on the other hand requires different optical and electrical descriptions. Optically, the 5 nm dispersed QD particles are much smaller than the wavelengths used in the simulation and are therefore represented by an effective dielectric function. Electrically, the composite active layer is treated as a homogeneous semiconductor, which is computationally convenient, but under such an approximation the localized QD states and any intermediate band cannot be reproduced. This surrogate is assigned a single effective bandgap of 1.44 eV, the volume-weighted average of the 1.03 eV dot and 1.55 eV host gaps at the 20 vol.% baseline fill ($0.20 \times 1.03 + 0.80 \times 1.55 = 1.446$ eV). The same value is retained across the fill-fraction sweep and in the 25 vol.% optimized case, where the volume-weighted average would instead be 1.42 eV. Recomputing the blend's above-gap fraction at that edge raises it from 0.591 to 0.605, which corresponds to a shift of order 0.3 percentage points in the strict single-gap efficiency and does not alter the comparison; the fill sweep

should nonetheless be read as holding the effective edge fixed while the dot content changes. Because this single edge stands in for two distinct absorbers, it is the parameter through which the collection assumption enters the embedded device, and its consequences are examined in Sec. V. The consequences of this approximation are examined through the two spectral interpretations described below.

All optical constants used in this work are experimentally derived. Measured complex refractive index data were used for MAPbI₃, MoO_x, FTO, SnO₂ and PbS-EDT, together with a set of measured PbS-QD spectra resolved both by dot size (3.5–5 nm) and by surface ligand and chemistry (oleic acid, EDT and NH₄I). The ligand-resolved dataset is what allows the quantum-dot bandgap sweep in Section IV to be treated as a materials comparison rather than a purely parametric one. Electrical parameters are taken from published measured or fitted tables for the same material set. The tabulated optical constants and the source of each electrical value are recorded in the accompanying repository.

B. Optical Simulation

The optical model was solved with the Ansys Lumerical FDTD solver [27]. Both designs were exposed to the same modelling setup. The structures were illuminated at normal incidence with the AM1.5G reference spectrum [26] over the range from 300 to 1700 nm, with periodic boundary conditions in the lateral directions and perfectly matched layer (PML) boundary conditions along the z axis. This wavelength window contains 945.62 W m^{-2} of the 1000.06 W m^{-2} total AM1.5G irradiance; efficiencies are nonetheless referenced to the full 100 mW cm^{-2} , so the 5.4% of incident power lying outside the simulated band is treated as unavailable rather than removed from the denominator. Photons below 300 nm are absorbed in the glass and FTO, and photons beyond 1700 nm lie below the bandgap of every absorber in the stack, so neither region contributes appreciable photocurrent.

The baseline unit cell is $50 \times 50 \text{ nm}$, expanded during the device optimization sweep to $600 \times 50 \text{ nm}$ to capture the effects of the textured rear contact. The rear grating has a 600 nm period with a 200 nm tooth height for D1 and 350 nm for D2. Optical simulations used mesh accuracy 2 for the baseline sweeps and mesh accuracy 3 for the converged optimized cells; the residual change in optical current between the two levels is below 2% (42.65 to 43.38 mA cm^{-2} for optimized D1). Absorbed optical power is converted into an electron-hole generation rate and imported into the electrical solver.

The ideal optical current is obtained from the absorbance as

$$J_{\text{opt}} = q \int A(\lambda) \frac{S(\lambda)\lambda}{hc} d\lambda$$

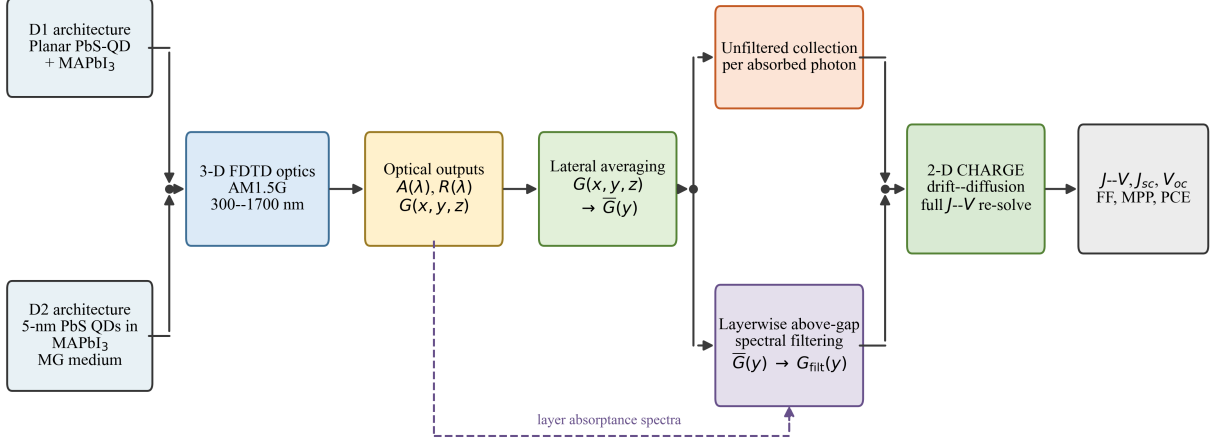


FIG. 2. Coupled simulation workflow, applied identically to both architectures. Three-dimensional FDTD optics over 300–1700 nm produce the absorptance $A(\lambda)$, reflectance $R(\lambda)$ and generation profile $G(x, y, z)$, laterally averaged to the depth profile $\bar{G}(y)$ supplied to the drift-diffusion solver. The profile enters CHARGE along one of two branches: unfiltered, crediting every absorbed photon as a collectable carrier, or layerwise above-gap filtering, scaling each absorber’s profile by its AM1.5G-weighted above-gap fraction before the electrical problem is re-solved in full. Both yield an independent J - V characteristic; their difference is the central comparison of this work.

where $A(\lambda)$ is the absorptance, $S(\lambda)$ the solar spectral irradiance, q the elementary charge, h Planck’s constant and c the speed of light. This quantity is an optical upper bound and is kept distinct from the short-circuit current J_{sc} returned by the electrical solver.

The dielectric function of the D2 composite active material was calculated using a Maxwell–Garnett mixing rule [23]. The rule is applicable in the sense that the 5 nm inclusions are far smaller than every wavelength simulated, so the blend is not optically resolved as a structured medium. Its accuracy at the fill fractions used here is not assumed on the grounds of diluteness, which at 20–25 vol.% would be difficult to justify, but is instead supported empirically by the comparison against explicit dot arrangements reported below. For dot volume fraction f ,

$$\beta = f \frac{\varepsilon_i - \varepsilon_h}{\varepsilon_i + 2\varepsilon_h}$$

$$\varepsilon_{\text{eff}} = \varepsilon_h \frac{1 + 2\beta}{1 - \beta}$$

where:

- ε_h is the MAPbI₃ host permittivity.
- ε_i is the 5 nm PbS-QD permittivity.
- $f = 0.20$ for the baseline.
- The effective spectrum is sampled and imported into Lumerical as a “Sampled 3D data” material.

To test the unified dielectric-function approximation, the 20% effective-medium result was compared against explicit square, hexagonal and honeycomb arrangements of 5 nm dots at matched fill. The effective medium yields an optical current of 44.51 mA cm^{−2} against 45.28–45.65 mA cm^{−2} for the explicit geometries, so the approximation underestimates the explicit-dot current by 0.77–1.14 mA cm^{−2}, roughly 2%. Separately, the three explicit packings span only 0.37 mA cm^{−2}, below the 0.59 mA cm^{−2} spread introduced by polarization alone; packing geometry is therefore unresolved at the present numerical accuracy and the three arrangements are treated as optically equivalent. The effective medium is an adequate optical simplification carrying a known small bias, but it is not evidence of electrical equivalence between a homogeneous blend and discrete dots. The optimized 25% fill is an extrapolation from the 20% comparison.

C. Electrical Simulation

The electrical model was solved with the Ansys Lumerical DEVICE (CHARGE) solver [27], using a two-dimensional depth cross section with a 50 nm width and a 10 nm out-of-plane normalization length, which converts the two-dimensional current to a per-unit-area value. All simulations were performed at 300 K. The optical model produces a three-dimensional generation dataset $G(x, y, z)$ in units of electron–hole pairs per cu-

bic metre per second. This profile is laterally averaged,

$$\overline{G}(z) = \frac{1}{L_x L_y} \iint G(x, y, z) dx dy,$$

and passed to CHARGE as a depth-dependent generation profile (Fig. 2).

Consequently the electrical simulation does not contain:

- Grating hot spots.
- Lateral current crowding.
- Tooth-corner field enhancement.
- Direct conformal metal/active interface effects.
- A fully textured electrical geometry.

The grating is therefore optical in FDTD but approximately flat and laterally averaged in CHARGE.

The electrical parameters used in the simulations are listed in Table II. The electrical mesh used a 1 nm minimum edge and a 20 nm maximum edge. A Newton solver was used to obtain the dark equilibrium solution before the illuminated simulations were run. The illuminated simulations used a voltage sweep from 0 to 1.0 V in 20 mV steps, extended to 1.4 V for the degenerately doped D2 points whose open-circuit voltage exceeds 1 V.

The MoO_x interlayer is resolved explicitly in the optical model but is not carried as a separate transport layer in CHARGE; it is instead represented by the work function of the combined Au/MoO_x anode, because the available standalone MoO₃ transport model prevented convergence. The FTO cathode and the Au/MoO_x anode were assigned work functions of 4.4 and 5.3 eV respectively and both were treated as ideal ohmic boundaries, so contact-barrier and contact-resistance losses lie outside the scope of this two-device comparison.

D. Spectral Filtering, Robustness and Reproducibility

The generation profiles imported from the optical simulation assume that every absorbed photon contributes one electron-hole pair, including photons with energies below the bandgap of the absorbing layer. The raw calculation therefore represents an unfiltered limit, in which all absorption is counted as generating collectable carriers. No mechanism for sub-bandgap collection is solved in this work; the unfiltered limit is the outcome that would be approached if such a channel, for example an operative intermediate band [1, 21, 24] or dot-to-host transfer [2], were fully effective. To obtain a strict single-gap result, each layer's generation profile was multiplied by its AM1.5G-weighted above-gap fraction, using bandgaps of 1.55 eV for MAPbI₃, 1.03 eV for the planar PbS-QD layer

and 1.44 eV for the D2 effective medium, with the corresponding cutoff wavelength taken as $\lambda_g = 1239.842/E_g$:

$$G_i^{\text{strict}}(y) = f_i^{>E_g} G_i^{\text{raw}}(y).$$

The filtered profiles were re-imported into CHARGE and the electrical problem was solved again in full, so the strict values are an independent solution rather than a rescaling of the raw result. This procedure is a form of parasitic-absorption accounting, standard in optical photovoltaic modelling; the variant applied here removes below-gap absorption inside the active layer rather than absorption in contact or metal layers. The two cases are referred to throughout as the unfiltered and strict single-gap limits. Both are model-defined, and neither is asserted as a physical bound on a real device.

The above-gap fraction $f_i^{>E_g}$ is a scalar computed from the layer's wavelength-resolved absorptance, so the filtered current integrated over each layer is exact; applying it as a uniform multiplier on $\overline{G}(y)$, however, is an approximation. Because sub-bandgap photons are weakly absorbed they penetrate further than above-gap photons, so the discarded generation is in reality weighted toward the rear of each absorber rather than distributed uniformly through it. The filtering therefore reproduces the correct layer-integrated above-gap current but an approximate depth profile within the layer. A wavelength-resolved treatment, in which $G(y, \lambda)$ is integrated only over above-gap wavelengths, would remove this approximation and is the natural refinement of the present workflow.

Robustness of the simulation assumptions was tested against interface surface-recombination velocity, D2 carrier mobility and doping, QD bandgap, grating geometry, QD fill fraction, temperature, illumination intensity, polarization, incidence angle, mesh accuracy, effective-medium formulation, and series resistance. Mesh and polarization studies are treated as numerical checks, while variations in recombination, mobility, doping, temperature, illumination and resistance are treated as physical parameter sensitivities. TE and TM comparisons differed by less than 1% in optical current.

Series resistance was applied after the CHARGE solution according to

$$V_{\text{ext}} = V - J R_s,$$

using $R_s = 0, 0.5, 1$, and $2 \Omega \text{ cm}^2$, followed by recalculation of the maximum-power point. This is a bounding sensitivity rather than a self-consistent sheet-resistance or current-crowding treatment.

All reported results are reproducible from the accompanying repository, which contains the simulation source models, materials data, builder scripts and analysis code.

E. Numerical and Solver Limitations

Two categories of unsuccessful configuration are distinguished in this work: those revealing a physical sensitivity of the device stack, which are reported with the

TABLE II. Electrical parameters used in the drift-diffusion model. Baseline and optimized values are listed separately where they differ; optimized transport values are best-case projections rather than typical measured values. All simulations were performed at 300 K.

Layer	E_g	EA (eV)	ϵ_r	Doping (cm^{-3})	Mobility μ_n / μ_p	SRH lifetime
PbS-EDT	1.14 eV	3.90 eV	20	$p \ 1 \times 10^{16}$	0.0002 / 20	10 ns
PbS-QD, D1	1.03 eV	3.85 eV	20	$n \ 1 \times 10^{15}$	0.02 / 0.02 baseline; 0.1 / 0.1 optimized	10 μs
MAPbI ₃	1.55 eV	3.90 eV	10	$p \ 1 \times 10^{15}$	10 / 10	100 ns baseline; 500 ns optimized
D2 blend surrogate	1.44 eV	3.92 eV	10	$p \ 1 \times 10^{15}$; 1×10^{16} high-voltage	10 / 10 or 20 / 20, depending on preset	5 μs
SnO ₂	3.44 eV	4.00 eV	9	$n \ 1 \times 10^{17}$	100 / 25	1 ns

corresponding results, and those arising from limitations of the implemented numerical model, which are collected here and excluded from the performance analysis.

The collapse of the tunnel-junction configuration is attributed to the absence of an explicit band-to-band tunneling treatment in the CHARGE model used, and is therefore not interpreted as evidence against the physical concept. Interface-recombination boundary conditions default to acting on minority carriers only; early scans configured this way produced no measurable effect until both carriers were enabled, and those early scans are excluded. A one-at-a-time scan of the blend effective bandgap over 1.40–1.50 eV returned an efficiency change of identically zero, which is inconsistent with the exponential dependence of the intrinsic carrier density, and hence of the open-circuit voltage, on bandgap. That scan is treated as not having reached the solver and is excluded from the sensitivity analysis of Fig. 11. The influence of the assumed collection edge is instead quantified optically above, through the change in the blend’s above-gap fraction when the edge is displaced. A conformal grating cross-section was constructed but could not be driven with a correctly distributed generation profile across the widened simulation domain, so all reported grating results use the laterally averaged flat geometry. Finally, optical mesh accuracy 4 could not be completed within the available compute budget; convergence is therefore reported at accuracy 3, where the residual change is below 2%. A fuller record of unsuccessful configurations is provided in the supplementary information.

III. RESULTS AND BASELINE COMPARISON

A. Baseline Comparison Under Conservative Parameters

Both architectures were first solved with the conservative parameter set of Table II, in which every transport value is taken from a published table rather than from the upper end of its reported range. This establishes a common starting point before any optimization is applied.

The planar device D1 reaches a short-circuit current of 39.07 mA cm^{-2} , an open-circuit voltage of 0.736 V and a fill factor of 0.716, giving a power conversion ef-

iciency of 20.59%. The embedded device D2 reaches 42.23 mA cm^{-2} , 0.715 V and 0.671, giving 20.27% (Fig. 3). The two efficiencies differ by 0.32 percentage points, but the underlying balance of the three figures of merit is markedly different. D2 delivers 3.16 mA cm^{-2} more current than D1, consistent with the broader near-infrared absorption of the blend seen in Fig. 4, while D1 recovers that deficit through a 21 mV higher open-circuit voltage and a fill factor 0.045 higher. The two morphologies therefore arrive at almost the same efficiency by opposite routes: D2 is absorption-strong, D1 is transport-strong.

The fill-factor difference is the larger of the two compensating terms and is the more informative. In D1 the photogenerated carriers are separated across a discrete MAPbI₃/QD heterojunction and collected through a continuous quantum-dot film with well-defined transport parameters. In D2 the same carriers must traverse a homogeneous blend whose effective mobility is lower than that of the host perovskite, which depresses the fill factor even though the absorbed photon flux is higher. This distinction becomes the organising theme of the optimization study in Section IV, where the two devices respond to entirely different levers.

It is important to separate the optical and electrical stages when reading these numbers. The generation profiles supplied to the electrical solver correspond to integrated optical currents of 39.49 mA cm^{-2} for D1 and 43.17 mA cm^{-2} for D2, so both devices collect close to, but below, their optical ceilings. Absorptance is an optical quantity and is not equivalent to external quantum efficiency; the conversion from one to the other is exactly what the drift-diffusion stage performs, and the small shortfall between J_{opt} and J_{sc} is the collection loss.

Two limits on what this baseline can support should be stated at the outset. First, the 0.32-point separation is not a morphology ranking: it is smaller than the spread produced later by plausible variation in single transport parameters, and it reverses under the optimized parameter set discussed in Section V. Second, both rows assume that every absorbed photon generates a collectable carrier, an assumption examined and relaxed in Section V. The baseline establishes that the two architectures are broadly competitive under matched, conservative assumptions; it does not rank either morphology.

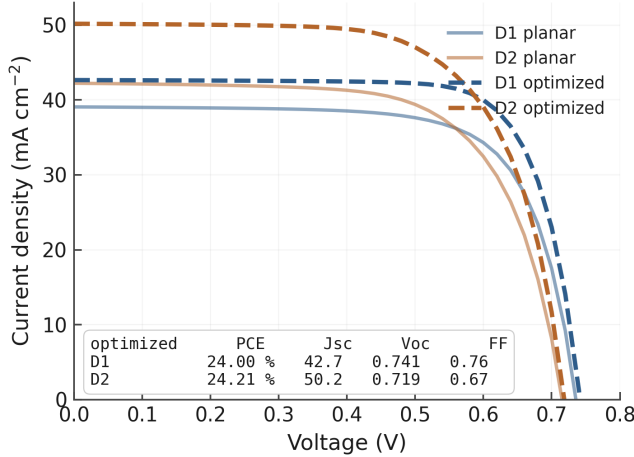


FIG. 3. Baseline and optimized current-voltage characteristics for the planar (D1) and embedded (D2) devices. Inset lists the optimized figures of merit.

B. Optical Response and Generation

Figure 4 compares the total absorbance and reflectance of the two architectures at normal incidence. Below the MAPbI₃ absorption edge at 800 nm, corresponding to the modelled 1.55 eV bandgap, the two devices are nearly indistinguishable: both absorb almost the entire incident flux, and the perovskite alone is sufficient to do so. The architectures separate beyond that edge. In this region absorption is carried entirely by the quantum-dot component, and the embedded blend sustains a consistently higher absorbance across the near infrared than the discrete planar film at matched active-layer thickness. Integrated over the full 300–1700 nm window, this yields optical currents of 39.49 mA cm⁻² for D1 and 43.17 mA cm⁻² for D2 at 20 vol.% dot fraction, and these are the generation profiles supplied to the drift-diffusion stage.

The difference is principally geometric. Both devices contain the same absorbing species, but D2 distributes the dots through the full 850 nm of the active layer, whereas D1 confines them to the lower 300 nm. Near-infrared photons that survive the perovskite reach dot material over a longer path in D2, and the blend therefore presents a larger optical cross-section to exactly the part of the spectrum the perovskite cannot use. That D2 does so while containing less quantum-dot material than D1 — 20 vol.% of an 850 nm layer is equivalent to a 170 nm film — indicates that distribution, not quantity, drives the gain. Two further differences nonetheless contribute and should not be attributed to geometry: the two cells carry different dot volumes, as quantified in Sec. V A, and the blend is represented by a Maxwell-Garnett effective medium whose absorption differs by roughly 2% from explicit dot arrangements at matched fill. Section V ex-

amines how much of the resulting advantage corresponds to above-bandgap absorption.

Both reflectance spectra show pronounced oscillations beyond roughly 900 nm, with peak-to-valley excursions approaching 0.4. These are Fabry-Pérot resonances of the thin absorber stack bounded by the specular gold rear contact: the optical thickness of the cell is comparable to the wavelength in this region, and the planar metal mirror preserves the phase relationship required for the interference to survive. The effect is not an artifact of the mesh, and it is present in both morphologies, since it originates in the cavity rather than in the absorber composition. It represents a real and recoverable loss, and its suppression is the subject of Section IV.

Finally, absorbance is an optical quantity and should not be read as external quantum efficiency. The conversion between them is performed by the drift-diffusion stage, and involves two distinct reductions: the collection loss between J_{opt} and J_{sc} , and the question of whether an absorbed photon was above the bandgap of the material that absorbed it. The first is a transport effect and is already contained in the baseline results of Section III A; the second is an assumption, and is examined separately in Section V.

C. Interface Robustness

The two morphologies differ in more than their optical response. The planar device separates carriers across a discrete MAPbI₃/QD heterojunction, whereas the embedded device has no discrete planar heterojunction; its quantum-dot/perovskite interfaces are instead distributed throughout the host. Because interface defects are a recognised limitation of perovskite devices, and local carrier lifetime is known to vary strongly with film microstructure [7], the tolerance of each architecture to interface recombination is a meaningful discriminator independent of the spectral question.

Figure 5 shows the efficiency of both devices as the surface-recombination velocity S applied at the outer active interfaces is swept from 0 to 10⁶ cm s⁻¹. The two architectures degrade at markedly different rates. The planar device falls from 20.59% to 14.58%, a loss of 6.0 points, whereas the embedded device falls from 20.27% to 17.23%, a loss of 3.0 points. The embedded cell is essentially insensitive up to 10³ cm s⁻¹ (20.27% to 20.23%) and overtakes the planar device already at $S = 10^2$ cm s⁻¹. Interface tolerance is therefore the clearest advantage the embedded morphology holds in this study, and it is an advantage of a different kind from the optical one: it survives the spectral filtering that removes the embedded device's apparent current benefit.

The internal heterojunction is the reason for the difference. Applying a representative $S = 100$ cm s⁻¹ at all three planar interfaces, including the internal MAPbI₃/QD junction that the sweep of Fig. 5 does not load, lowers every planar row by roughly three points:

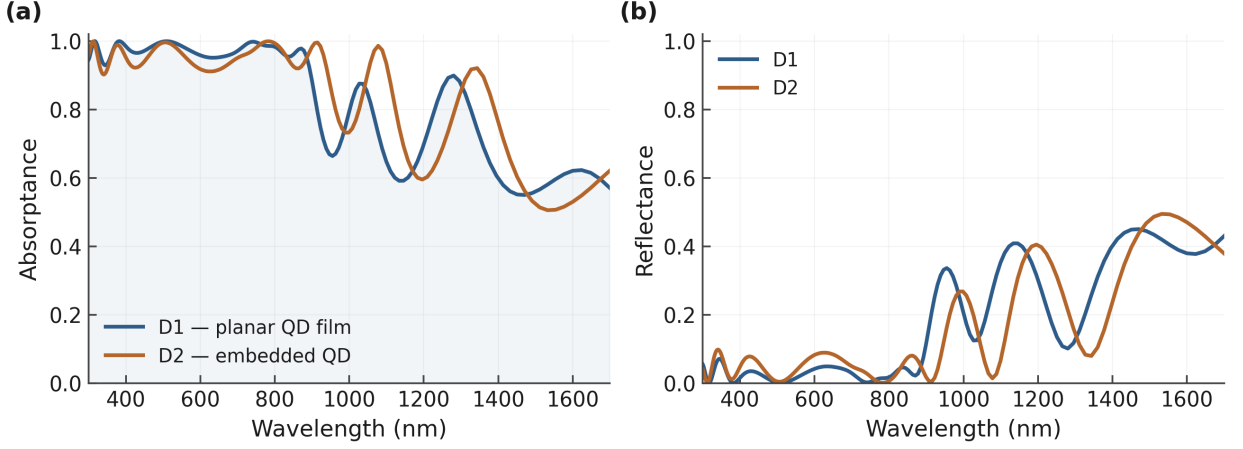


FIG. 4. Optical response of the two architectures at normal incidence. (a) Total absorbance and (b) reflectance over 300–1700 nm. Both devices absorb almost completely below the MAPbI₃ cutoff; beyond it the embedded blend sustains higher absorbance across the near infrared, while the interference structure visible in both channels originates from the thin absorber combined with a specular rear reflector.

the conservative baseline falls from 20.59% to 17.00% and the optimized cell from 24.00% to 20.00%, with the corresponding strict single-gap values falling from 16.20% to 13.50% and from 18.80% to 15.74%. The strict single-gap lead over the embedded device then narrows from 4.57 points to 1.51 points. It does not reverse, but its magnitude clearly depends on interface quality, and the headline rows of Table I should be read as the defect-free limit rather than as a projection for a fabricated cell. A per-interface decomposition of this loss is given in Section VI.

This tolerance requires one qualification. The recombination swept here acts on the planar interfaces that the model resolves; it does not represent the quantum-dot/perovskite interface distributed throughout the blend. At 20 vol.% and 5 nm diameter that internal area amounts to roughly $2 \times 10^2 \text{ m}^2$ per m^2 of device, an interface class the homogeneous effective medium cannot describe, since its carriers never encounter a dot boundary. This is not a claim that the planar device has only one interface: its quantum-dot film is itself a nanocrystal solid whose inter-dot and ligand interfaces are folded into the measured transport parameters used here [5]. The distinction is that the planar device’s dominant heterojunction is explicitly modelled and swept, whereas the embedded device’s dominant internal interface is not represented at all. Its apparent robustness is therefore conditional on that interface being benign, which the present model cannot verify and which remains the principal open question for the embedded architecture.

A separate scan established that this stack is sensitive to band alignment as well as to interface quality. Raising the electron affinity of the PbS-EDT layer from 3.90 to 4.40 eV collapses the short-circuit current from roughly 37 to 7 mA cm^{-2} . At the higher affinity the layer no longer presents a barrier to electrons arriving

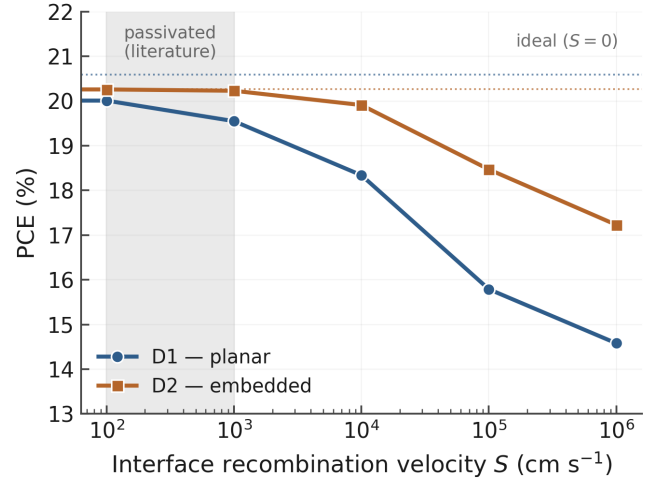


FIG. 5. Power conversion efficiency as a function of interface recombination velocity for the planar and embedded morphologies. The shaded band marks the range reported for passivated lead-halide perovskite surfaces [25].

from the absorber and begins to extract them at the anode, opposing the intended hole-selective function and short-circuiting the device internally. The magnitude of the collapse reflects the loss of carrier selectivity rather than a numerical instability, and it identifies the hole-transport-layer affinity as a parameter that must be controlled to within a few hundred meV for either morphology to operate at all.

IV. OPTIMIZATION AND DESIGN LEVERS

Device optimization was conducted through a staged set of deterministic parameter studies. The optical per-

formance was first optimized in the FDTD solver with the aim of higher optical current generation; the selected generation profiles were then transferred to the electrical solver and optimized for conversion efficiency. The resulting optimized devices should therefore be read as the best cases within the tested parameter grids rather than as globally optimal designs.

A. QD Bandgap and Spectral Design

The QD material was selected from six PbS datasets spanning bandgaps from 0.80 to 1.24 eV (Fig. 6). The best device performance, 20.38%, occurred with the 1.03 eV NH_4I -passivated QD, with $J_{\text{sc}} = 38.49 \text{ mA cm}^{-2}$, $V_{\text{oc}} = 0.737 \text{ V}$ and $FF = 0.719$. Lowering the bandgap to 0.80 eV increased J_{sc} to 41.69 mA cm^{-2} , but caused V_{oc} to decline to 0.568 V, which led to a reduced efficiency of 14.46%. Conversely, the 1.24 eV material recovered V_{oc} but sacrificed near-infrared current. The selected 1.03 eV material therefore represents the best current-voltage trade-off among the sampled materials. Because the optical datasets include both NH_4I - and oleic-acid-capped QDs, this comparison is more accurately described as a materials screen than as a pure bandgap sweep. This screen was run with a slightly different parameter set from the baseline device of Sec. III A, so its optimum of 20.38% is not directly comparable to the 20.59% quoted there; only the ordering across QD materials is used. Not every material in the optical dataset was viable electrically. Oleic-acid-capped dots were excluded before the electrical stage: although their optical constants are acceptable, the long insulating oleate ligand limits inter-dot transport to a mobility of order $10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ [16], low enough that the film is effectively non-conducting. Only the shorter-ligand EDT and NH_4I chemistries were carried into the device comparison.

B. Grating and Light Trapping

Texturing the rear contact produced a substantial optical enhancement (Fig. 7). For D1, replacing the flat Au rear layer with a 600 nm-period, 200 nm-tooth grating increased J_{sc} from 40.03 to 42.14 mA cm^{-2} and the PCE from 21.14% to 22.34%. D2 showed a larger response: a 600 nm-period, 350 nm-tooth grating increased efficiency from 19.88% to 22.30%. The magnitude of the gain depends on period, tooth height and polarization, and these values are quoted from the grating map at their stated configuration labels rather than merged with the separately optimized final devices.

The interaction between texturing and dot loading could not be resolved with the present data. The fill sweep of Fig. 8 was performed with a flat rear contact and peaks at 20 vol.% (20.27%), whereas the optimized grating configuration uses 25 vol.%. No fill sweep was carried out with the textured contact, so the 25 vol.% value is

a configuration choice for the optimized case rather than a swept optimum under the grating, and the two cannot be compared directly.

Finally, the grating enters the electrical model only through a laterally averaged generation profile, as described in Sec. II A. The textured geometry is therefore optical in FDTD but flat in CHARGE, so current crowding, tooth-corner field enhancement and conformal-contact effects are excluded from the reported grating efficiencies.

C. Embedded-Device Transport and Doping

The D2 device was strongly sensitive to the transport properties assigned to the composite active layer (Fig. 8). Increasing the composite mobility from 0.02 to $50 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ raised the fill factor from 0.576 to 0.678 and the unfiltered PCE from 21.94% to 24.30%, while V_{oc} decreased from 0.827 to 0.714 V. A mobility of $20 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ was used for the optimized comparison, producing 24.21%. Changing the p-type doping produced a second trade-off between voltage and current: at 10^{16} cm^{-3} the PCE reached 24.91%, while the highest sampled unfiltered point reached 27.50% at 10^{17} cm^{-3} . These mobility and high-doping values are upper-envelope sensitivities rather than evidence that a fabricated hybrid film attains them, and the 27.50% point is excluded from the morphology comparison because no strict single-gap counterpart has been computed for it. Depth grading did not improve D2: both tested grading directions reduced the efficiency below the 20.27% uniform reference, which is consistent with D2 being absorption-limited rather than collection-limited in this model.

Combining the optimized optical and electrical parameters raised the unfiltered efficiency from 20.59% to 24.00% for D1 and from 20.27% to 24.21% for D2. Applying strict single-gap filtering to the same optimized configurations gives 18.80% for D1 and 14.23% for D2. The planar device responded to bulk transport differently. Raising the D1 dot mobility from 0.02 to $0.1 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ left the fill factor unchanged, and raising the dot doping from 10^{15} to 10^{19} cm^{-3} produced no change at all. In the tested configuration the planar device is therefore limited by its interfaces rather than by bulk transport in the quantum-dot film, so improvements to the film's bulk properties do not translate into device gains; this conclusion is specific to the architecture examined here.

V. SPECTRAL CONSISTENCY AND THE TWO-REGIME COMPARISON

The comparison developed in the preceding sections rests on one assumption that the optical model cannot resolve on its own: whether a photon absorbed below the electrical bandgap of the material that absorbs it

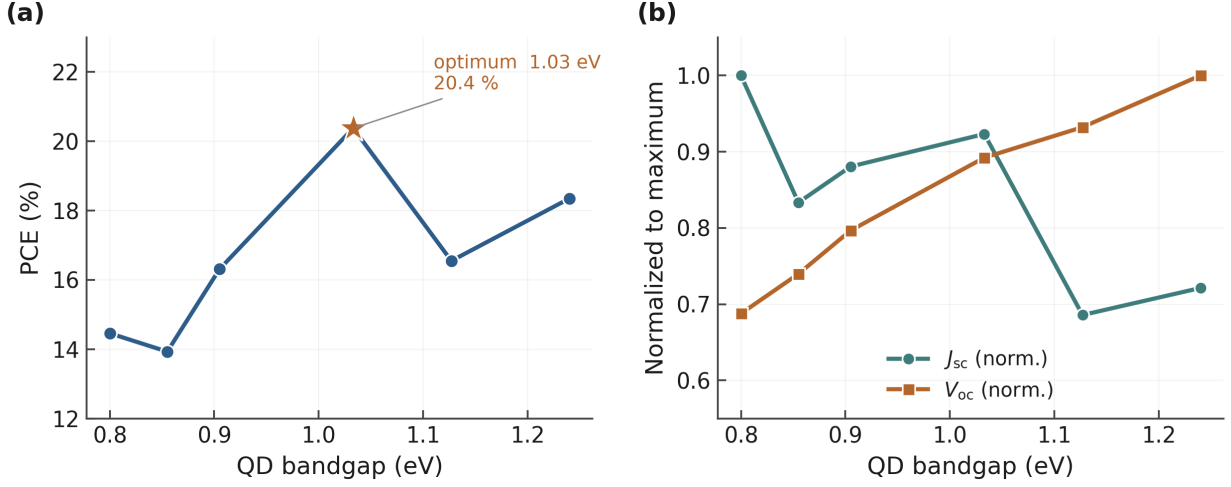


FIG. 6. Quantum-dot bandgap sweep. (a) Power conversion efficiency against QD bandgap, showing the optimum at 1.03 eV. (b) Short-circuit current and open-circuit voltage normalized to their maxima, showing the current–voltage trade-off across the sampled range.

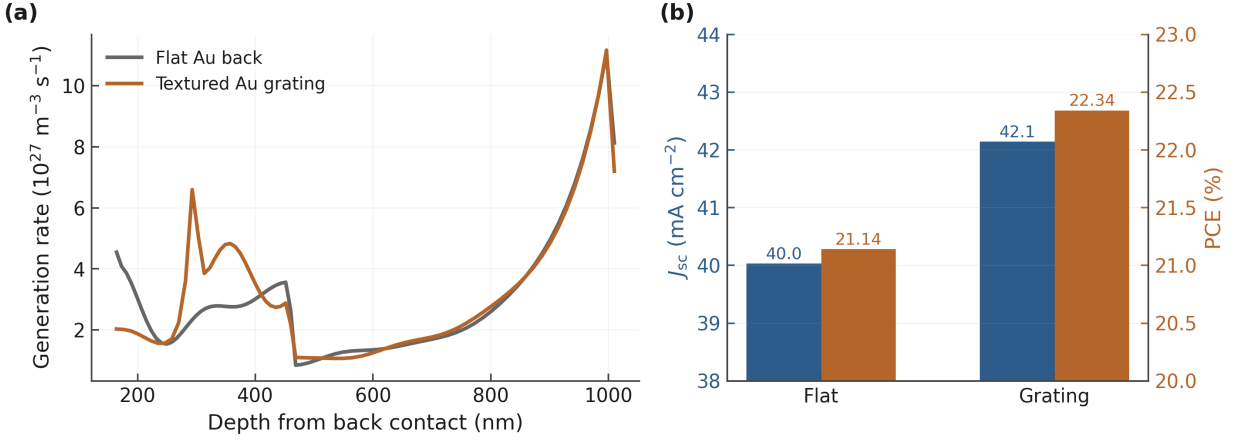


FIG. 7. Effect of the textured rear contact. (a) Depth-resolved generation rate for a flat Au reflector and for the 600 nm-period grating. (b) Resulting short-circuit current and power conversion efficiency.

produces a collectable carrier. This section makes that assumption explicit and shows that it, rather than any single headline efficiency, decides which morphology is preferred. The two limits defined in the methodology are carried throughout: the unfiltered limit, in which every absorbed photon contributes to generation, and the strict single-gap limit, in which each absorber’s generation is restricted to its above-gap fraction and the drift-diffusion problem is solved again in full.

One qualification applies to both limits. The comparison is not between two placements of an otherwise identical material: it sets a continuous 1.03 eV PbS-QD film against a homogeneous 1.44 eV effective-medium surrogate for the blend. The two absorbers therefore differ in collection edge as well as in geometry, and the strict single-gap result compares those two representations rather than isolating morphology alone.

The optical energy budget frames the result. After

spectral filtering the planar device retains 79.4% of its integrated optical current, whereas the embedded device retains only 59.1%; the blend carries 18.33 mA cm^{-2} of below-gap generation that the single-gap reading discards. The embedded morphology’s optical advantage (Fig. 4) is disproportionately near-infrared absorption lying below the collection edge, and whether that absorption is credited as photocurrent decides the comparison.

Under the unfiltered limit the two optimized devices are indistinguishable, at 24.00% (D1) and 24.21% (D2). The 0.21-point separation is smaller than the residual numerical uncertainty (below 2% in optical current between mesh accuracy 2 and 3) and must be read as parity rather than an embedded-device advantage. This reading is also unstable: the ranking it produces reverses between the conservative baseline and the optimized parameter set (Table I), so it cannot by itself carry a conclusion about morphology.

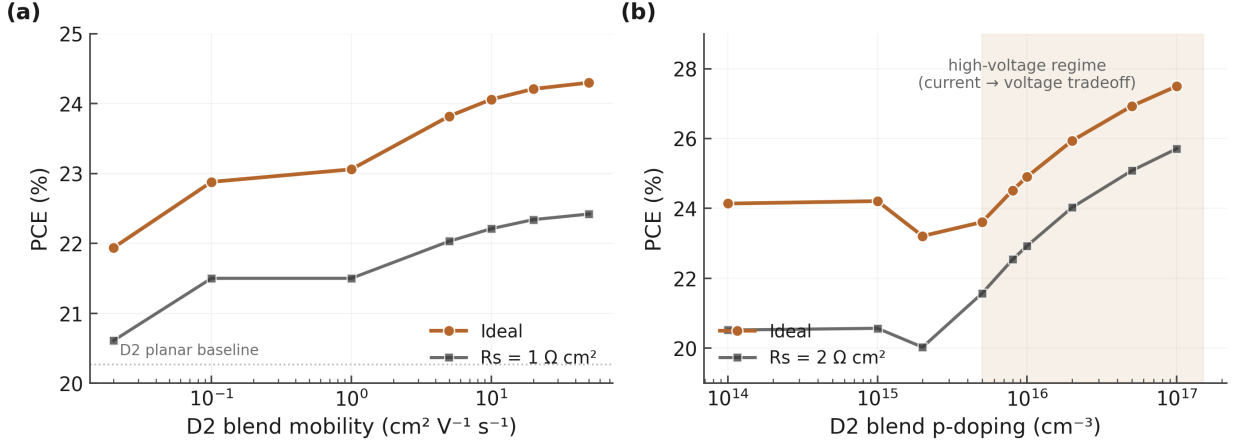


FIG. 8. Embedded-device transport sensitivity. (a) Efficiency against blend carrier mobility. (b) Efficiency against blend p-type doping, including the high-voltage regime in which current is traded for open-circuit voltage.

The strict single-gap limit is decisive and, unlike the unfiltered limit, points the same way at every parameter set. The optimized planar device leads by 4.57 percentage points (18.80% versus 14.23%), and the conservative baseline leads by a comparable margin (16.20% versus 11.81%). These strict values are not rescaled efficiencies: the filtered generation profiles were re-imported into CHARGE and the full electrical problem solved again, so they carry the same transport, recombination, and contact physics as the raw solutions. The layer-resolved absorption budget (Fig. 9) and the side-by-side comparison of the two limits (Fig. 10) show the origin of the gap and its consequence for efficiency.

A. Matched Quantum-Dot Loading

The planar and embedded devices compared above do not carry the same quantity of quantum-dot material: the planar film is 300 nm of continuous PbS, whereas the embedded blend holds 20 vol.% dots across an 850 nm layer, equivalent to a 170 nm QD thickness. Placement is therefore not the only variable that distinguishes them. To separate placement from loading, we built planar devices whose quantum-dot volume is matched to the embedded cell while the total active thickness is held at 850 nm: a 170 nm film, equal to the 20 vol.% blend, and a 212.5 nm film, equal to the 25 vol.% blend, with the perovskite thickness increased to compensate. Each was solved under both limits with the conservative parameter set (Table III).

At matched loading the strict single-gap ranking is unchanged. The planar device at 170 nm QD reaches 15.73% against the embedded cell's 11.81%, a 3.9-point lead at identical quantum-dot volume, and the 212.5 nm film reaches 16.84%. The planar advantage under the strict reading is therefore not an artifact of its larger dot loading. Two further observations reinforce the point. First, thinning the planar film from 300 to 170 nm

TABLE III. Matched quantum-dot-loading control. Planar (D1) devices whose quantum-dot volume equals the embedded (D2) cell at 20 and 25 vol.%, with the 850 nm active thickness held fixed, solved under both limits with the conservative parameter set. The standard 300 nm planar device and the 20 vol.% embedded device are listed for reference.

Device	QD film	Unfiltered	Strict
		PCE (%)	PCE (%)
Planar, standard	300 nm	20.59	16.20
Planar, matched 20 vol.%	170 nm	20.47	15.73
Planar, matched 25 vol.%	212.5 nm	21.58	16.84
Embedded, 20 vol.%	—	20.27	11.81

costs only 0.5 points under the strict reading (16.20% to 15.73%), because that reading already discards the sub-gap near-infrared absorption that the additional dot material mainly provides. Second, exchanging dot material for perovskite is mildly favourable when sub-gap collection is inoperative: the 212.5 nm matched film slightly exceeds the standard 300 nm planar device. Under the unfiltered limit the matched planar devices remain at parity with the embedded cell (20.47–21.58% versus 20.27%). This control is drawn at the flat, conservative level, with the quantum-dot volume matched at fixed total active thickness, and is not a re-optimization of either device.

B. Interpretation and Limits

The mechanism behind these results is straightforward. The embedded morphology absorbs more near-infrared light because its dots are distributed through the full depth of the absorber, but in a homogeneous single-gap medium that absorption sits below the collection edge and generates no carriers. The embedded cell becomes competitive only if a genuine sub-bandgap collection channel is present, whether by intermediate-band transitions, dot-to-host transfer, or percolation through the dot phase; none of these can be represented by a ho-

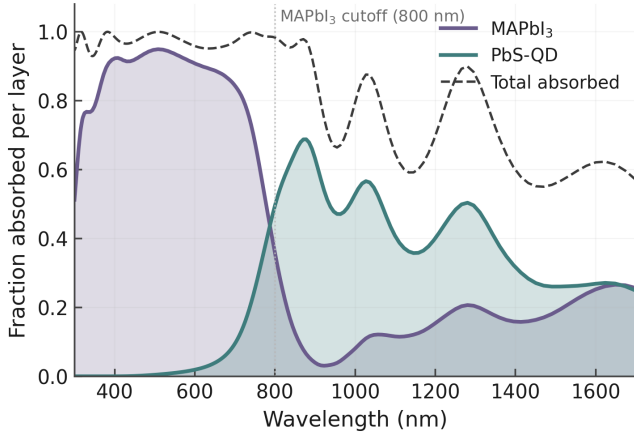


FIG. 9. Layer-resolved optical absorption budget for D1 and the MAPbI₃ absorption cutoff at 800 nm. Absorptance is an optical quantity and is not equivalent to external quantum efficiency.

homogeneous effective medium. The strict single-gap result is therefore best read as the conservative limit, which applies when no such channel operates, and the unfiltered result as the optimistic limit, which applies when one operates perfectly. Neither is a physical bound on a real device; both are model-defined scenarios that bracket the behaviour a real device could show.

The headline comparisons assume defect-free interfaces ($S = 0$); as shown in Sec. III C, a representative interface velocity narrows the strict single-gap lead without reversing it. Separately, the high-voltage embedded point in Table I (24.91% unfiltered) is a distinct operating regime reached by heavy p-type doping of the blend, which trades current for open-circuit voltage; it is not a variant of the optimized morphology used in the two-limit comparison, and its strict single-gap counterpart (14.78%) does not change the ranking.

Two caveats apply to these results. The comparison established here is between collection *limits*, and between the two absorber representations under those limits; the model does not itself solve the sub-band transport of a discrete quantum dot, and so cannot predict which limit a real embedded device would approach. What it does establish is that the planar morphology is preferred under the conventional single-gap picture and merely at parity under the most optimistic collection assumption. Sub-bandgap photocurrent has been reported in PbS-in-perovskite systems [1, 2], but not for a homogeneous blend at the 20–25 vol.% loading and film geometry modelled here, and it is that specific demonstration the embedded architecture still requires.

VI. SENSITIVITY AND LIMITATIONS

The following results are to be considered as perturbations in the numerical simulations and should not be

interpreted as measured device performance or the device’s confidence interval or statistical uncertainty. One parameter was changed at a time while all other optical and electrical parameters were held constant unless mentioned otherwise. This approach helps identify the dominant local dependencies, but it does not capture the effects of parameter interactions.

A. One-at-a-Time Sensitivity

In the D1 device, the widest efficiency interval was produced by the QD electron affinity, 18.48–22.43%, which demonstrates the necessity of band alignment at the discrete MAPbI₃/QD heterojunction. The MAPbI₃ lifetime and the QD minority-carrier lifetime produced PCE spans of 19.43–21.79% and 18.10–20.96% respectively. Increasing the interface recombination velocity from 0 to 10^4 cm s^{-1} reduced the PCE from 20.59% to 18.34%, while changes in the SnO₂ electron affinity produced a 19.21–20.95% range. All of these spans indicate that the D1 response is primarily controlled by the band alignment, lifetime and recombination properties of the junctions rather than by optical absorption alone.

For the D2 one-at-a-time sweeps, the electron affinity of the SnO₂ layer resulted in the largest PCE range, 17.04–20.37%, while the corresponding interface recombination sweep changed the PCE from 20.27% to 19.91%. The composite active layer lifetime produced only a 20.25–20.27% variation around this reference state. Separate D2 transport sweeps reveal a strong bulk transport–voltage trade-off: increasing the composite layer mobility from 0.02 to $50 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ increased the fill factor from 0.576 to 0.678 and the PCE from 21.94% to 24.30%, while V_{oc} decreased from approximately 0.827 to 0.714 V. Following the same pattern, the unfiltered doping sweep increases the voltage while reducing the current at the same time. All D2 blend-bandgap trials over 1.40–1.50 eV returned zero-valued outputs. That scan is consequently excluded from quantitative interpretation and should not be treated as evidence of bandgap insensitivity.

B. Series Resistance

Series resistance was introduced as a lumped post-processing sensitivity using

$$V_{\text{ext}} = V - JR_s, \quad (1)$$

where V and J are obtained from the resistance-free device solution. This approach therefore acts as a modifier to the terminal power curve but does not directly affect the internal electric field, recombination or carrier distribution. For the unfiltered optimized cases, increasing R_s from 0 to $2 \text{ } \Omega \text{ cm}^2$ reduced the D1 PCE from 24.00% to 20.94% and the D2 PCE from 24.21% to 20.57%. The small initial advantage of D2 is lost because its

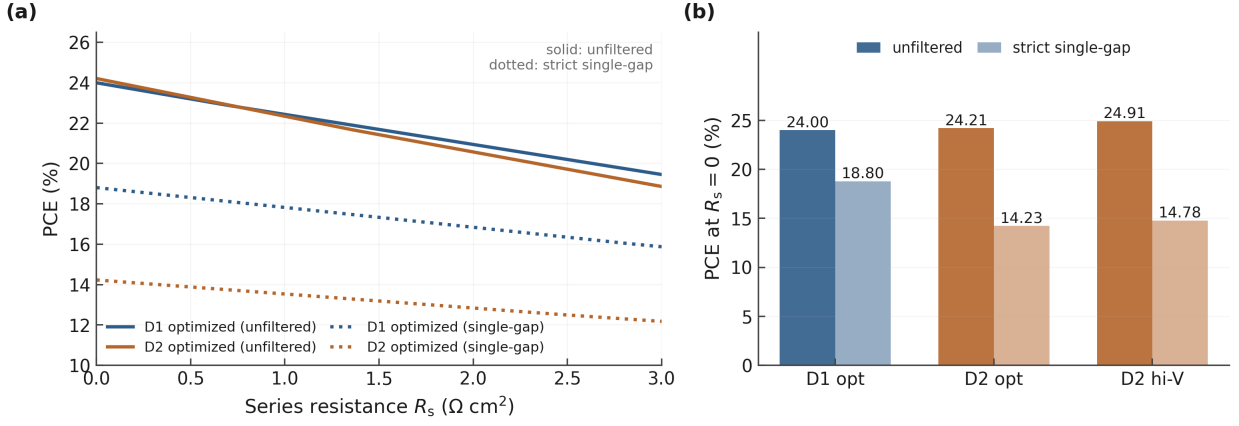


FIG. 10. Comparison of the unfiltered and strict single-gap limits. (a) Efficiency against lumped series resistance for both devices under both limits. (b) Efficiency at zero series resistance, including the separate high-voltage D2 operating point.

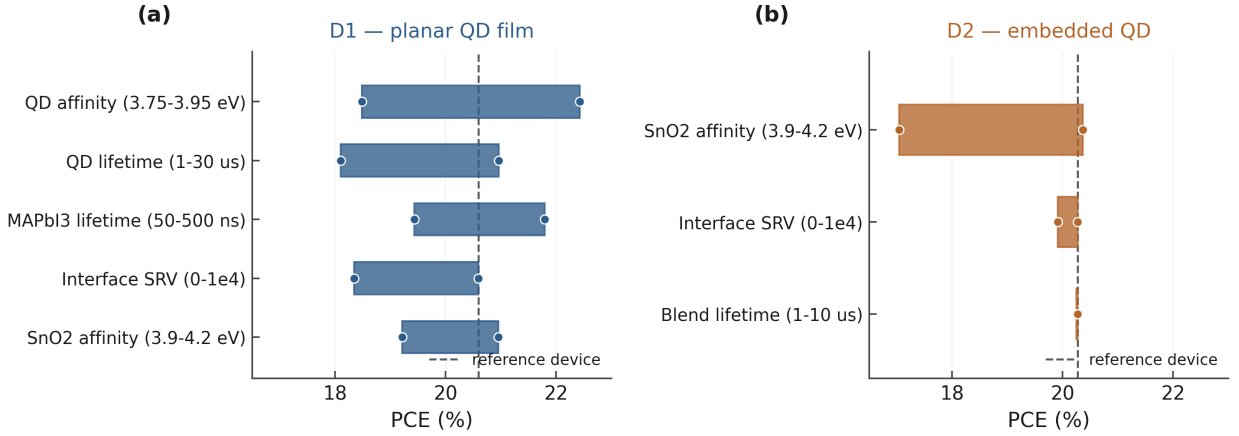


FIG. 11. One-at-a-time parameter sensitivity for (a) the planar and (b) the embedded device. Bars span the efficiency range produced by varying each parameter alone about its reference value; the dashed line marks the reference device.

higher current produces a larger resistive penalty, giving a crossover near $R_s \approx 0.76 \Omega \text{ cm}^2$ (Fig. 12). The separate high-voltage D2 sensitivity point decreased from 24.91% to 22.93% over the same range. Under strict single-gap filtering, D1 remains ahead across the tested resistance range; at $R_s = 0$, the corresponding efficiencies are 18.80% for D1 and 14.23% for D2.

C. Per-Interface Recombination

The D1 interface map in Fig. 13 applies a recombination velocity to one interface at a time while all other interfaces are held at $S = 0$. The dominant loss channel is the internal MAPbI₃/QD junction: at $S = 10^6 \text{ cm s}^{-1}$ its PCE decreases from 20.59% to 8.26%. Applying the same velocity to the other interfaces, QD/EDT or SnO₂/MAPbI₃, produces a smaller PCE reduction, which confirms that the additional internal heterojunction is the main recombination liability of the planar architecture under severe interface loss. In

Fig. 13(b), the electron-only calculations leave the complementary carrier velocity at zero while the symmetric calculations assign the velocity to both carrier channels. Their difference at the SnO₂/MAPbI₃ boundary shows the asymmetric carrier response of the drift-diffusion interface condition. It should not be interpreted as a direct model of a particular defect chemistry or capture cross section. The main devices assume ideal interfaces with $S = 0$. When all three explicit D1 interfaces are assigned $S = 100 \text{ cm s}^{-1}$, the optimized unfiltered PCE decreases from 24.00% to 20.00%, while the strict single-gap PCE decreases from 18.80% to 15.74%. The strict D1 advantage over the corresponding D2 reference consequently narrows from 4.57 to 1.51 absolute percentage points. The planar advantage is therefore retained at this finite-SRV baseline, but its magnitude depends strongly on the achievable passivation of the discrete QD interfaces.

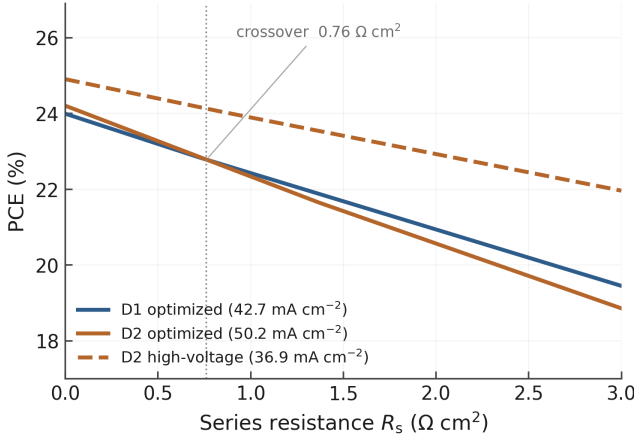


FIG. 12. Efficiency against lumped series resistance for the three unfiltered operating points, showing the D1/D2 crossover near $0.76 \Omega \text{ cm}^2$.

D. Numerical and Physical Limitations

The optical simulation is mesh size dependent. When increasing the FDTD mesh accuracy, the optically generated current increased from 42.65 to 43.38 mA cm^{-2} for optimized D1, a 1.7% change, and from 49.90 to 50.57 mA cm^{-2} for optimized D2, a 1.3% change. These residual changes quantify optical-source sensitivity only and should not be interpreted as a complete uncertainty interval for PCE. The composite active layer of the D2 device is modelled using a Maxwell–Garnett effective medium. At 20% QD fraction, this approximation produced 44.51 mA cm^{-2} compared with 45.28 – 45.65 mA cm^{-2} for the explicit 5-nm QD patterns, corresponding to an approximately 2% optical difference. The choice of mixing rule is not itself a significant source of uncertainty: an alternative Bruggeman effective medium [22] agreed with Maxwell–Garnett to within 1.7% in the complex index at 25% fill. Its use at the 25% fraction of the optimized grating case is nonetheless an extrapolation beyond the explicit 20% pattern check. The FDTD generation profile is laterally averaged before being imported into the drift-diffusion solver. This approach retains the depth-dependent generation but removes the lateral grating hot spots, current crowding and nanoscale QD connectivity. The resulting model therefore compares layer-averaged architectures rather than providing a fully coupled three-dimensional electro-optical prediction.

VII. DISCUSSION

The result of this study is best stated as a conditional rather than a ranking. Which morphology is preferred is decided by the collection assumption, not by geometry alone: the two architectures are at parity when all absorption is credited, and the planar cell leads by 4.57 per-

centage points when only above-gap absorption is. That ordering is not an artifact of the planar cell carrying more quantum-dot material, since it survives at matched dot volume (Sec. V A). What the study establishes, therefore, is that the embedded morphology’s optical advantage is contingent on a collection channel that conventional single-gap transport does not provide.

The robustness of the two architectures should not be read as a symmetric trade-off. Each is robust along a different axis, but the two claims do not carry equal weight. The planar cell’s advantage under spectral filtering is consistent across the tested parameter sets: it follows from where the absorption sits relative to the collection edge, and it persists at the conservative baseline, at the optimized point, and at matched dot loading. It remains conditional on the assumed blend collection edge and on the approximate depth profile of the filtering. The embedded cell’s tolerance to interface recombination is conditional, because the recombination swept in Sec. III C acts only on the planar interfaces the model resolves. Its dominant internal interface, the quantum-dot/perovskite boundary distributed through the blend, is absent from a homogeneous effective medium and therefore cannot degrade in these sweeps. The honest statement is thus narrower than “planar is never worse”: the planar morphology is preferred under the conventional single-gap picture and at parity under the most optimistic one, while the embedded morphology is more tolerant of the specific interfaces this model represents, and untested at the ones it does not.

That asymmetry has a practical counterpart. A discrete planar film presents a single, well-defined heterojunction, which is an engineering problem with an established toolkit and a measurable figure of merit in the surface recombination velocity. Dispersing the dots removes that junction but replaces it with quantities that are harder to measure and to control: the validity of the effective-medium description, the transport of a blend whose mobility depends on inter-dot coupling and ligand chemistry [5, 16], and the presence or absence of sub-bandgap collection. The embedded route is not disadvantaged in principle; it is disadvantaged in what can presently be verified about it.

Placing these numbers against the literature requires keeping four categories distinct. Certified device efficiencies for standalone PbS-QD cells [8, 9] are measurements of fabricated devices. Experimental PbS-in-perovskite hybrids [3] are measurements of the embedded morphology itself, but at loadings three orders of magnitude below the one modelled here. One-dimensional drift-diffusion projections [13, 14] report higher efficiencies than either, but without resolving the full-wave optical response that distinguishes the two morphologies in this work. The values reported here belong to a fourth category: coupled three-dimensional optical and two-dimensional electrical simulations of idealised contacts and best-case transport. They are not comparable to certified efficiencies and are not offered as predictions of

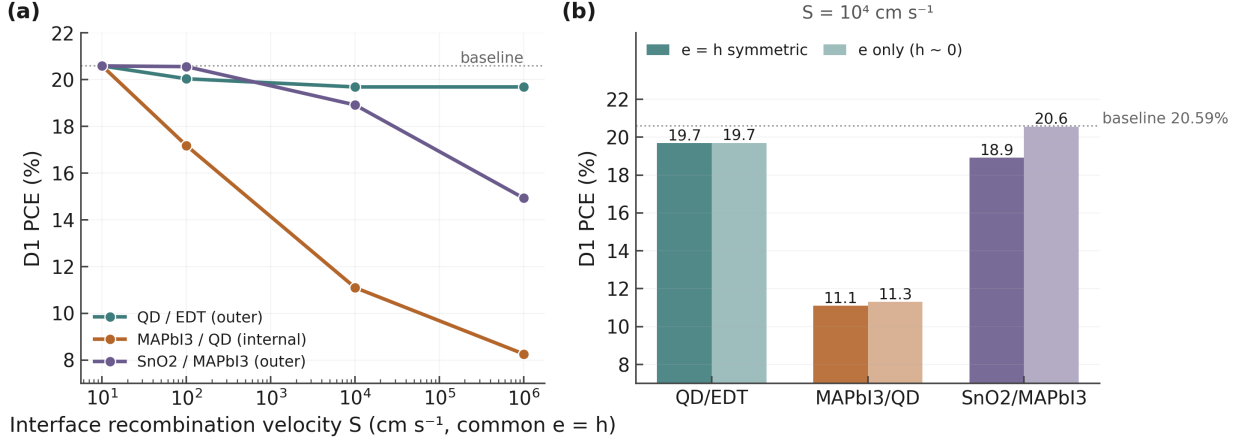


FIG. 13. Per-interface recombination diagnostic for the planar device. (a) Efficiency against recombination velocity applied at each interface in turn. (b) Efficiency at $S = 10^4$ cm s⁻¹ for symmetric and electron-only recombination, isolating the carrier-asymmetric response of the SnO₂/MAPbI₃ interface.

a fabricated cell.

The most important limitation is where the experimental evidence stops. Han’s measurement supports the strict single-gap limit as a physical statement, but, as Sec. VII A shows, at their dilute loading the simulation cannot discriminate between the two limits at all: the dots add so little absorption that crediting or discarding it changes the predicted current by less than the numerical resolution of the model. The discriminating evidence at that loading is experimental, not computational. At the 20–25 vol.% loading where the present claims are made, no comparable measurement exists, and the dot phase is dense enough that inter-dot coupling and percolation become plausible [16]. The conclusions of this work are therefore model-internal in exactly the regime that matters most.

Four measurements would settle what the model cannot. The first is sub-bandgap external quantum efficiency on a blend at working loading, which directly distinguishes the two limits. The second is the blend’s effective collection edge, which the present model must assume rather than derive. Recomputing the above-gap fraction of the simulated blend absorptance with that edge displaced across the range spanned by the host and dot gaps (1.55 to 1.03 eV) changes the retained fraction of optical current from 0.52 to 0.85. This is an optical calculation rather than a solved device sweep, but the span is wide enough that the assumed edge is among the most consequential inputs to the strict single-gap comparison. The third is direct evidence of dot-to-host carrier transfer or percolation, which determines whether a dot-localised carrier is collectable at all. The fourth is the interface recombination velocity at the quantum-dot/perovskite boundary, which the present model omits entirely and on which the embedded morphology’s apparent robustness depends. Blend mobility and current-crowding behaviour under a textured contact remain secondary but would tighten the optimized comparison.

A. Consistency with an Experimental Hybrid Device

The central assumption of the strict single-gap interpretation is that near-infrared photons absorbed below the host bandgap are not collected as photocurrent. This assumption can be checked directly against the one experimental study that reports both absorption and spectral response for a PbS-in-MAPbI₃ hybrid film, namely Han *et al.* [3]. In that work 4 nm PbS dots were dispersed in a MAPbI₃ absorber at 0.5 mg mL⁻¹ (corresponding to a dot volume fraction of roughly 3×10^{-4} , nearly three orders of magnitude below the 20 vol.% studied here). The hybrid film absorbs slightly further into the near infrared, to about 900 nm, than the pure perovskite; however, the reported incident-photon-to-current efficiency shows no significant contribution beyond ~820 nm, and the measured current gain originates almost entirely in the 400–800 nm range. The authors attribute the improvement to enhanced perovskite crystallisation, carrier mobility and fill factor rather than to infrared harvesting.

To test whether the present workflow reproduces this behaviour, the same coupled optical model was applied to a MAPbI₃ absorber containing 4 nm PbS dots at Han’s loading, using the measured 4 nm PbS optical constants. The active-layer absorptance and its above-gap fraction (referenced to the 1.55 eV, 800 nm MAPbI₃ edge) are reported in Fig. 14 and Table IV. At Han’s loading the dots increase the active-layer optical current by only 0.03 mA cm⁻², and the above-gap fraction is essentially unchanged from the pure film (0.735 versus 0.734): the small optical addition the dots make sits almost entirely on the long-wavelength side of the collection edge. In fact the above-gap (collectable) current is unchanged to within 0.01 mA cm⁻², so the entire addition falls below the gap. Under the strict single-gap interpretation this addition is discarded, so the predicted current gain from the dots is negligible — consistent with the absence

TABLE IV. Absorber-level consistency check against Han *et al.* [3]. Optical current and above-gap fraction (800 nm MAPbI₃ edge) for an 850 nm MAPbI₃ active layer seeded with 4 nm PbS dots at the experimental loading, with pure perovskite and the 20 vol.% blend as anchors.

Case	Dot vol. fraction	J_{opt} (mA cm ⁻²)	Above-gap fraction
Pure MAPbI ₃	0	32.59	0.735
Han, 0.5 mg mL ⁻¹	3.3×10^{-4}	32.62	0.734
Han, 1 mg mL ⁻¹	6.6×10^{-4}	32.65	0.734
This work, 20 vol.%	0.20	42.34	0.562

of sub-820 nm response in Han’s measured IPCE. The unfiltered limit, by contrast, would credit the full near-infrared absorption as collectable current, which the experiment does not support.

The scope of this check must be stated carefully, because it is narrower than it may first appear. The mesoporous TiO₂/spiro-OMeTAD architecture and 490 nm absorber thickness of the experimental cell are not modelled here, and the comparison is made at the level of the absorber’s spectral response. More importantly, at Han’s loading the two limits are not distinguishable by simulation: the dots add only 0.03 mA cm⁻² of optical current, so whether that addition is credited or discarded changes the predicted current by an amount below the numerical resolution of the model. The discriminating evidence at this loading is therefore experimental rather than computational — it is Han’s own contrast between an absorption edge extending to ~900 nm and an IPCE that is flat beyond ~820 nm that indicates sub-bandgap absorption is not collected in that film.

Two consequences follow. First, this comparison supports the strict single-gap limit as a physical statement about dilute PbS-in-MAPbI₃ films, but it does not validate the model at the 20–25 vol.% loading used elsewhere in this work, where the dot phase is dense enough for inter-dot coupling, percolation, or mini-band formation to become plausible and where no comparable measurement exists. Second, the effective-medium description itself is not confirmed at this loading: Han report a measurable extension of the absorption edge, whereas the Maxwell–Garnett blend at 3×10^{-4} predicts a change too small to resolve, which suggests that the experimental enhancement arises from a mechanism outside the model, such as dot aggregation or a modification of the perovskite itself. Extending the fill fraction toward 20 vol.% drives the above-gap fraction down to 0.562, quantifying how much of the embedded morphology’s apparent optical advantage would be forfeited if sub-gap collection is inoperative there, but that extrapolation rests on the model rather than on this experiment.

B. Future Modeling Path

The gap between the two limits is, at root, a modelling gap. The Maxwell–Garnett description averages

the 5 nm dots and the host into a single medium with one effective bandgap, so the electrical solver cannot ask whether a photon absorbed in the near infrared was above the gap of a discrete dot or below the gap of the host. Resolving that question is the natural next step, and it requires an electrical model of the dot phase rather than a better optical one.

The concrete route is an explicit dot geometry in the drift-diffusion solver: a 1.03 eV quantum dot embedded in the 1.55 eV perovskite matrix, with a carrier-escape or thermionic-emission treatment governing whether a dot-localised carrier reaches the host and is collected [20]. The optical side of this already exists, since the square, hexagonal and honeycomb 5 nm patterns were built for the effective-medium check; what is missing is the electrical transport and carrier-transfer model. Such a model would place the embedded device somewhere between the two limits bracketed here rather than leaving it undetermined, and would connect the present accounting to the intermediate-band picture [21, 24] that a homogeneous medium cannot represent.

The same explicit-dot framework is the natural setting for surface-passivation chemistry, which the present model does not address. Core-shell dots such as PbS/CdS or PbS/ZnS are a standard route to suppressing trap-assisted recombination and tuning band alignment [18, 19], and a reviewer may reasonably ask why they are not considered here. The reason is that the comparison in this work is one of spectral accounting: it turns on where absorption sits relative to the collection edge, whereas passivation acts on carrier lifetime and band alignment and does not move photons across that edge. Those two channels are, moreover, already spanned by the sensitivity analysis: the quantum-dot lifetime (1–30 μ s, 18.10–20.96%) and electron affinity (3.75–3.95 eV, 18.48–22.43%) sweeps of Sec. VI bracket the range a shell would access, and neither reorders the two morphologies. Passivation is therefore not expected to change the present conclusion. It nonetheless belongs to the future path for two reasons specific to the embedded device. First, the homogeneous blend has no dot surface to passivate, so the effect of a shell on the distributed quantum-dot/perovskite interface — the interface this model omits — can only be examined once that interface is represented explicitly. Second, a wide-gap shell raises the barrier for a carrier to escape from the dot into the host, which would push the embedded device *toward* the strict single-gap limit treated here as the conservative bound, rather than away from it.

Finally, the two limits bracket the one experimental hybrid efficiency available for this system (18.6% [3]), so reproducing that device with an explicit-dot electrical model would be a direct validation: a converged result falling inside the bracket would confirm the transport treatment, while one falling outside it would indicate a missing channel. Where a lower-cost estimate is needed before that model is built, an optically active deep-level or two-step absorption term could stand in for the dot,

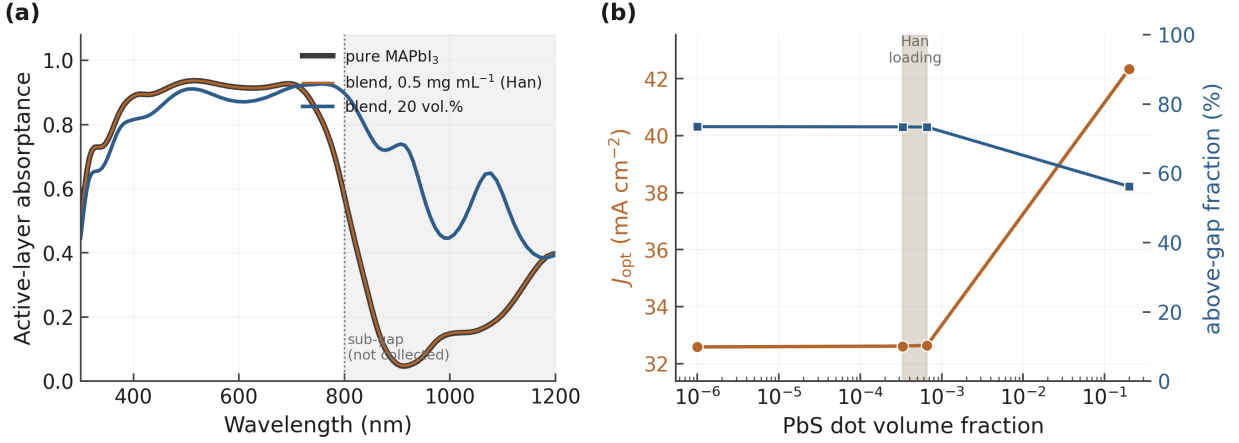


FIG. 14. Absorber-level validation against Han *et al.* [3]. (a) Active-layer absorbance for pure MAPbI₃, the Han-loading blend, and the 20 vol.% blend. Pure and the Han loading are indistinguishable, the dots being negligible at 3×10^{-4} , while the 20 vol.% blend adds absorption almost entirely beyond the 800 nm collection edge (shaded, not collected under the single-gap reading). (b) Optical current and its above-gap fraction versus dot fill, from Han’s loading (shaded) to the 20 vol.% used elsewhere in this work.

but it should be understood as less faithful than an explicit dot with a solved escape process.

VIII. CONCLUSION

We compared planar and embedded PbS quantum-dot integration in a common perovskite stack using coupled three-dimensional FDTD optics and two-dimensional drift-diffusion transport. Under the unfiltered limit, in which every absorbed photon is credited as photocurrent, the optimized planar and embedded devices are at parity, at 24.00% and 24.21%. Under the strict single-gap limit, in which only above-gap absorption contributes, the planar device leads by 4.57 percentage points, 18.80% against 14.23%. The ranking is preserved when the quantum-dot volume of the two cells is matched at equal active thickness, so it is not an artifact of loading.

The practical reading is that planar integration is the more robust route whenever the model is required to remain consistent with conventional single-gap transport, whereas the embedded morphology becomes competitive only if a sub-bandgap collection channel, by intermediate-band transitions, dot-to-host transfer, or percolation, is experimentally established at working loading. These conclusions are bounded by the model’s principal simplifications: a homogeneous single-gap surrogate for the blend, an approximate depth profile in the spectral filtering, laterally averaged grating generation, ideal ohmic contacts, and best-case optimized transport values. The decisive next step is experimental rather than computational: a direct measurement of sub-bandgap collection, together with the quantum-dot/perovskite interface recombination velocity, in matched planar and embedded devices.

IX. DATA AND CODE AVAILABILITY

All source models, materials data, and analysis code required to reproduce the results are available in the accompanying public repository at <https://github.com/mohamedelsayed-0/qd-study>. These comprise the FDTD (FSP) and CHARGE (LDEV) device models, the measured optical constants and the electrical parameter tables, the builder and sweep scripts, the figure-generation code, and the CSV, JSON, and MAT result files behind each figure and table. The headline device rows of Table I reproduce from named presets in the preset runner, for example `d1_max` and `d2_max` together with their spectrally filtered counterparts. Full parameter tables, the complete sensitivity sweeps, the mesh, polarization and incidence-angle checks, the effective-medium comparison, and secondary figures are provided in the Supplementary Information. The FSP and LDEV containers are supplied as binary solver artifacts rather than plain-text model definitions; the builder scripts that generate them are included so that each model can be re-generated from source.

ACKNOWLEDGMENTS

The authors thank Dr. Haitham Omran for facilitating access to Ansys Lumerical FDTD and CHARGE through the GUC university network.

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